

Design of Detection and Assessment System for Electric Vehicle Charging Station Based on Genetic Algorithm

Weixin Liu^{1,a}, Xiaowen Wang^{2,b}, Xiaolei Zhang^{1,c}, Yang Ding^{1,d}, Qingtian Li^{3,e}

¹ State Grid Xinjiang Company Limited Electric Power Research Institute, Wulumuqi, Xinjiang, 830000 China

² Xinjiang Uygur Autonomous Research Institute Of Measurement and Testing, Urumqi, Xinjiang, 830011 China

³ State Grid Aletai Electric Power Supply Company, Aletai, Xinjiang, 836500 China

^a1557453864@qq.com, ^b370153891@qq.com, ^ctomme2235@163.com, ^d978110224@qq.com, ^e659462760@qq.com

Abstract—With the large-scale electric vehicle (EV) charging load connected to the power grid, EV charging stations have been widely used. Basic charging facilities are supporting facilities to ensure the convenience of EV travel and affect the growth of EV industry. When building a charging station, several performance indexes with different dimensions should be considered when selecting a charging station. At present, a large quantity of EV charging stations have been built and operated all over the country, and most of them are unattended. In order to maintain the normal operation and use of the charging station, great attention should be paid to the detection and operation maintenance. In this article, aiming at the abnormal state in the detection of DC charging station, the phenomenon and reasons are analyzed, and the detection and assessment model of EV charging station based on genetic algorithm optimized neural network (GANN) is proposed, and the operation state of charging station is analyzed to complete the fault diagnosis. The simulation results show that the assessment model is highly robust and the assessment results are more objective, so that the operation and maintenance strategy of charging stations can be put forward, the performance and safety problems of charging equipment can be solved, and the sustainable growth of EV industry can be promoted.

Keywords—Genetic algorithm, Electric vehicles, Charging station, Fault detect

I. INTRODUCTION

Due to the rapid growth of human society, energy crisis and environmental problems are becoming more and more serious, and energy conservation and emission reduction has become a new challenge for all countries[1]. Improving the efficiency of energy utilization, developing and utilizing renewable energy, strengthening the protection of ecological environment and realizing sustainable development have become the common goals of the growth of human society. Under this goal, various renewable new energy sources have emerged. In the field of energy, replacing conventional fossil energy with electric energy can reduce the use of fossil energy and reduce greenhouse gas emissions[2]. The construction of EV infrastructure is necessary and urgent. In the EV industry, the charging station is a very critical and important part, and the state pays great attention to it[3]. In recent years, a quantity of guiding documents have been issued to promote the better development and faster construction of charging facilities. However, whether EV can be popularized lies in that it needs to protect the environment and save energy while maximizing the

convenience of users. The basic charging facilities play a vital role, which is a supporting facility to ensure the convenience of EV travel and affect the growth of EV industry[4]. In order to maintain the normal operation and use of the charging station, great attention should be paid to the detection and operation maintenance. Therefore, it is necessary to carry out in-depth analysis and research on the detection technology of EV charging station, and strengthen the product specification to help the industry develop better[5]. Due to the rapid growth of EV industry and charging technology, the performance and safety of charging equipment have been paid more and more attention. In the traditional fault diagnosis, manual diagnosis is used, but because the charging stations are widely distributed, managers can't find the fault in time, which has great security risks and will cause a lot of labor cost waste[6]. Mohammed et al. proposed using non-dominated sequencing GA to optimize the designed EV charging and demand-side power grid, so as to reduce the operating load of the power grid and improve the overall operation safety and energy-saving effect. However, due to the lack of online detection method, it is impossible to detect the charging station in real time[7]. Donna et al. proposed to use the fault dictionary method to check the electrical components in the charging station. By analyzing the structural characteristics of each electrical component in the equipment, the fault of the charging station can be effectively detected. However, due to the wide distribution of equipment, it is necessary to arrange manpower to regularly check each design, which wastes a lot of labor costs[8]. In this article, aiming at the abnormal state in the detection of DC charging station, the phenomenon and reasons are analyzed, and the detection and assessment model of EV charging station based on GANN is put forward, and the operation state of charging station is analyzed to complete the fault diagnosis.

II. METHODOLOGY

A. Overall Scheme Design of Charging Station

Relevant departments should pay more attention to the on-site inspection of charging station equipment, reduce the possibility and probability of dangerous factors, and carry out professional maintenance of charging station equipment to ensure that the charging station will not be affected by any factors during use, improve the safety of EV and power grid operation, and provide help and support for the better growth of EV charging technology. The whole power quality detection work should be carried out in a relatively stable state of current, which is helpful to ensure that the

data obtained after detection has stronger reliability and accuracy[9]. The charging station should be able to charge quickly, and ensure the quality of output electric energy to avoid charging damage of EV and vehicle battery. The charging station sends user information, charging state parameters, fault information and other data to the remote management system through the wireless network module; The remote management system can send commands to the charging station actively, and realize remote start-stop control, parameter modification, parameter reading and other functions[10]. The whole detection system has the characteristics of intelligence and automation, and some detection devices can be connected independently to expand their functions, thus achieving the goal of carrying out load adjustment according to the rated power of the charging station. The adjustable DC load can be resistive load or electronic load, and the load size is controlled according to the detection process to achieve the function of adjusting the output voltage or current of the charging station.

Near the power supply of AC charging station, inspectors usually need to determine the direction of actual work to ensure that the pile head is selected to connect to the power supply when the charging station is closed. The load controller and other devices are comprehensively used to accurately adjust the voltage, current and other parameters in the detection to obtain the specific information of the charging state. At this time, the current value and voltage value should be recorded in detail, and the time of testing work should also be included[11]. In order to meet the peak power demand, the transmission and distribution network can be upgraded to increase the power level, but this will lead to overcapacity in the normal operation. In addition, there are potential risks in charging at night, and the pricing mechanism designed to prevent daytime charging will overload the distribution transformer designed for overnight cooling. Once the inspector puts forward the instruction request, all kinds of information

generated in the test will be sent and stored efficiently in the test simulation device. After the charging station is adjusted to a constant current state, the inspectors should immediately adjust the output current to make the setting value as the rated value, and input the information about the rated voltage, so as to maintain the state that the voltage setting value can change within the required range, and then conduct a secondary test on the input and output active power of the charging station, and save the final result. When the fault is eliminated, if the power supply voltage meets the charging requirements at this time, normal operation can be resumed as soon as possible.

B. EV Charging Station Detection and Assessment Model

When the detection work is unfolded, the professional staff need to connect the charger chain with the load, then set its current and voltage to ensure that it is in a constant voltage state, then input the rated voltage, and constantly adjust the load current to reach the rated current input value, and carry out the measurement work according to each module[12]. In this scheme, firstly, a large amount of information data is collected by designing an operation state acquisition device on the charging platform, and the operation state acquisition device is operated at the charging equipment terminal, and the collected data is divided into real-time operation data and historical state data of the equipment. The energy storage device is mainly responsible for storing the excess output power of the photovoltaic power generation system. When the required power of the EV load in the charging station is less than the output power of the photovoltaic power generation system, the energy storage device is started to store energy, and then the energy is released when the power demand on the load side increases. The control center collects and monitors state variables such as voltage, current and power, and according to the collected information, each system controller executes its control objectives. The GA process of EV charging station detection and assessment is shown in figure 1.

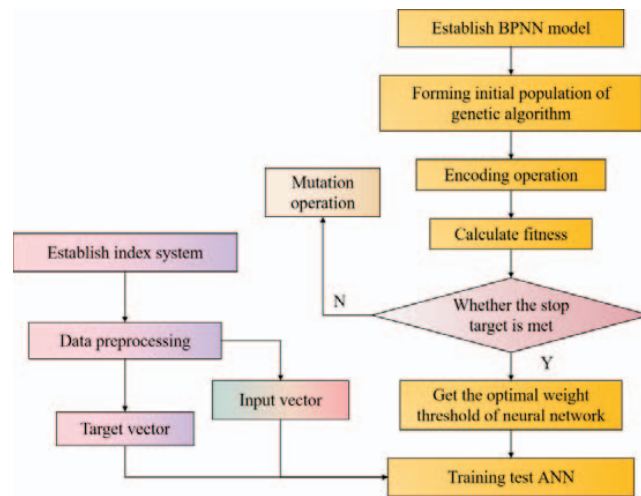


Figure 1 GA process of EV charging station detection and assessment

In the real-time operation data, the charging equipment is detected online in real time during the charging process, and the equipment is detected in the charged state, and in the historical state data of the equipment, it is detected by manual detection in the off-line state of the equipment[13]. In order to ensure that the EV can be more stable and safe

during operation, it is necessary for professional staff to conduct a comprehensive inspection of the functional operation status of the EV charging station, find out the problems at the first time, and then formulate perfect methods to properly solve the problems. By analyzing the design defects of equipment through historical fault data,

the operation process of power grid is optimized, and the influence of surrounding weather data on the operation of equipment is analyzed, and all the collected data are stored in the database to complete the collection of equipment operation data.

Assuming that there are n EV in the area, divide the day into J segments at the time interval Δt , and establish the EV charging state set according to the user's travel rules:

$$C_j(i) = \begin{cases} 1, & i \in T_{v,j} \\ 0, & \text{other} \end{cases} \quad (1)$$

Where $C_j(i)$ is the charging state of the j EV at the i time. When $C_j(i) = 1$, the j EV can be charged at the i time. When $C_j(i) = 0$, the j EV is out at the CC time.

Assuming that the power required by the i EV on that day is Q_j , the quantity of charging stations is m and the rated charging power is $P_{e\max}$, in order to meet the charging demand of EV, the following formula should be satisfied:

$$\sum_{i=1}^J P_e(i) = \sum_{j=1}^n Q_j \quad (2)$$

$$N = \min \left[m, \sum_{j=1}^n C_j(i) \right] \quad (3)$$

Where: $P_e(i)$ is the sum of the charging powers of all EV at the i moment; N takes the minimum value of the quantity of EV that can be charged and discharged in the charging station and the quantity of charging stations.

When the network side is overloaded, the control strategy of the charging station will be triggered. In this case, the output power of the hybrid power supply side is constant, and the power fluctuation of the load side will be adjusted by the connected EV in proportion through the designed charging algorithm. In the process of data storage in the database, the data is processed, the basic parameters of each device are arranged in time sequence by using linked list, and the running States of the devices are connected in series through interactive data and environmental data, so as to facilitate the next use of data. Whether the problems and faults during charging appear early warning signals in time to reflect the charging operation state. Check the input function of charging station. During the power-on process of the charging station, it is necessary to simulate the startup state to check whether the charging station is in normal operation.

In the process of GA evolution, by introducing an information exchange mechanism, information can be transmitted between two populations, which helps

individuals avoid wrong information judgment and fall into local optimum. Nonlinear dynamic adaptive inertia weight strategy is adopted to improve the performance of the algorithm. Its update status is as follows:

$$w(t) = w_{end} + (w_{start} - w_{end}) \times \exp \left(-k \times \left(\frac{t}{t_{max}} \right)^2 \right) \quad (4)$$

Among them, k is the control factor; Control the smoothness of w and t curves. Calculate the total path index:

$$S_k = \frac{1}{F} \left(\frac{n+1}{h_y \cdot j} \right) \quad (5)$$

Where S_k represents the distance between the k distribution points; $1/F$ represents the distance returned after the delivery is completed; $h_y \cdot j$ is the adjustment coefficient of the total path; n is the total path index value.

III. RESULT ANALYSIS AND DISCUSSION

EV charging station detection and assessment system mainly uses the collected data of charging station to analyze and realize the assessment of the relevant health status of charging station, so as to form a targeted operation and maintenance strategy to guide the operation and maintenance personnel. When a new EV is connected to the charging station and the network side is overloaded, in order to reduce the pressure on the power grid, the self-adjustment algorithm of EV is activated. The charging EV reduces its output power proportionally to solve the problem of system power fluctuation caused by accessing a new EV. Figure 2 shows the comparison between the model output value and the actual value.

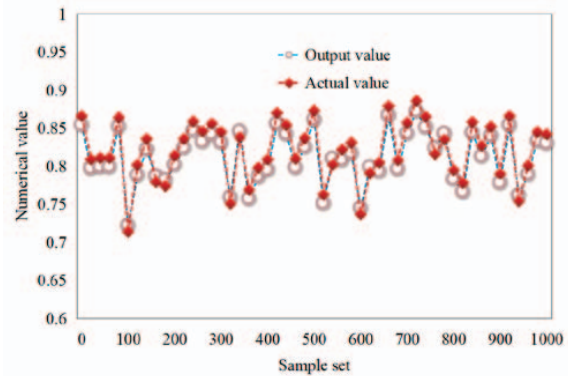


Figure 2 Comparison between model output value and actual value

When the monitoring module and EV charging station upload data to the monitoring platform synchronously according to the unified communication protocol, their data are compared and calculated by the software system, and the calculated data is compared with the monitoring threshold. The data comparison result can be used as the analysis basis

for remote monitoring of urban EV charging stations. In order to better verify the optimization ability of the optimization algorithm in solving the model, different algorithms are used to carry out charging station detection experiments, and the optimal iteration times of the algorithm after each execution are obtained. The comparison of fault detection accuracy of the algorithm is shown in Figure 3.

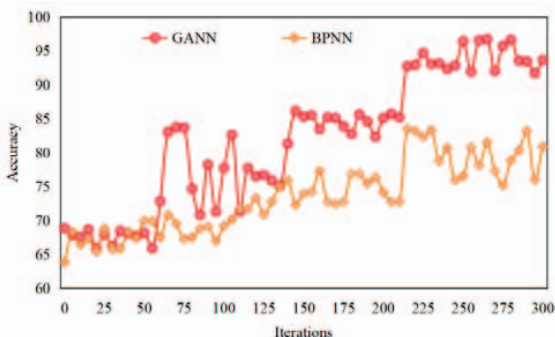


Figure 3 Comparison of fault detection accuracy of algorithms

It can be seen that compared with the other two algorithms, the accuracy of GANN is higher, and the accuracy of the algorithm reaches the best after 200 iterations. The detection and assessment method of EV charging station based on GANN has good adaptive ability, and can infer the relationship between model input and output through learning and training according to the provided charging station state data, thus completing efficient detection and assessment of charging station state. If the communication of the detector is overtime during the normal charging operation of the charging station, the charging activity will be terminated, and at the same time, it will enter the state of reconnection for the third time, and the phenomenon of restarting charging will also occur. After the charging is stopped, the position of the contactor in the low-voltage auxiliary power supply circuit will also lead to the phenomenon that the disconnection time is incorrect. It is necessary for inspectors to disconnect the contactor that does not meet the standard to meet the specific needs of putting into the discharge circuit.

Through the charging station management system, the collected charging station operation data and maintenance data are obtained, and the big data are analyzed and processed to assess the health status of charging station equipment, so as to formulate the charging station operation and maintenance strategy, so as to guide the growth of on-site operation and maintenance work and improve the operation and maintenance efficiency. The comparison of the search efficiency of the algorithm is shown in Figure 4.

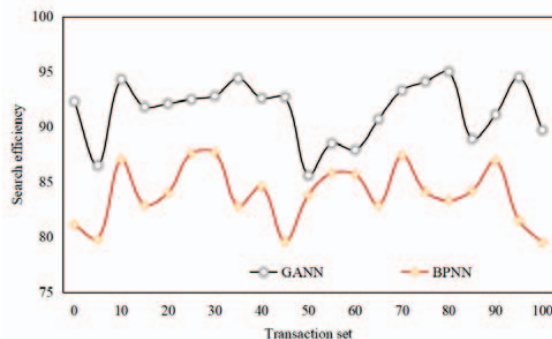


Figure 4 Comparison of search efficiency of the algorithm

It can be seen that detecting the state of charging station based on GANN has obvious advantages in operation efficiency. The algorithm needs to calculate, assess and select the fitness function value in the solution space of the problem, while the chromosome genetic operation is carried out in the code space of the algorithm. The charging station detection technology will be added to EV industry. This technology will give full play to its own functions, and it can find out the input and output problems of charging station current and voltage in time, ensure that it is always in a stable and safe state during operation, and ensure that the charging station's electric energy application is safer. The successful research on remote monitoring method of EV charging station will greatly reduce the cost of industrial measurement, improve the supervision technology of government charging station product quality, improve the efficiency of government supervision of EV charging station, and promote the healthy and orderly growth of EV charging station industry.

IV. CONCLUSIONS

The increase in the quantity of EVs will put greater pressure on the heavily-loaded power grid, pose new challenges to the distribution network, and have an impact and impact on the power quality of the network side. In order to maintain the normal operation and use of the charging station, great attention should be paid to the detection and operation maintenance. Therefore, it is necessary to carry out in-depth analysis and research on the detection technology of EV charging station, and strengthen the product specification to help the industry develop better. In this article, aiming at the abnormal state in the detection of DC charging station, the phenomenon and reasons are analyzed, and the detection and assessment model of EV charging station based on GANN is put forward, and the operation state of charging station is analyzed to complete the fault diagnosis. Detecting the state of charging station based on GANN has obvious advantages in operation efficiency. The detection and assessment method of EV charging station based on GANN has good adaptive ability, and can infer the relationship between model input and output through learning and training according to the provided charging station state data, thus completing efficient detection and assessment of charging station state. Through the designed control strategy, the EV charged in the charging station can share the power fluctuation of the system in proportion, without affecting the output power of the AC system in the hybrid power supply system, thus reducing the pressure on the power grid and alleviating the

heavy load pressure.

REFERENCES

- [1] Lam A Y S, Leung Y, Chu X. Electric Vehicle Charging Station Placement: Formulation, Complexity, and Solutions. *IEEE Transactions on Smart Grid*, vol. 5, no. 6, pp. 2846-2856, 2017.
- [2] Monteiro V, Lima P, Sousa T, et al. An Off-Board Multi-Functional Electric Vehicle Charging Station for Smart Homes: Analysis and Experimental Validation. *Energies*, vol. 13, no. 8, pp. 1864, 2020.
- [3] Li Y, Huang Y, Zhang M. Short-Term Load Forecasting for Electric Vehicle Charging Station Based on Niche Immunity Lion Algorithm and Convolutional Neural Network. *Energies*, vol. 11, no. 5, pp. 1253, 2018.
- [4] Zhong Y, Xia M, Chiang H. Electric vehicle charging station microgrid providing unified power quality conditioner support to local power distribution networks. *International Transactions on Electrical Energy Systems*, vol. 27, no. 3, pp. 1-13, 2017.
- [5] Mohammed A S, Anuj, Lodhi A S, et al. Techno-economic feasibility of hydrogen based electric vehicle charging station: A case study. *International journal of energy research*, vol. 2022, no. 10, pp. 46, 2022.
- [6] T Donna, et al. Locating Electric Vehicle Charging Stations: Parking-Based Assignment Method for Seattle, Washington. *Transportation Research Record*, vol. 2385, no. 1, pp. 28-36, 2018.
- [7] Xu X, Niu D, Li Y, et al. Optimal Pricing Strategy of Electric Vehicle Charging Station for Promoting Green Behavior Based on Time and Space Dimensions. *Journal of Advanced Transportation*, vol. 2020, no. 12, pp. 1-16, 2020.
- [8] Fathabadi H. Novel solar powered electric vehicle charging station with the capability of vehicle-to-grid. *Solar Energy*, vol. 142, no. 1, pp. 136-143, 2017.
- [9] Balasundar C, Sundarabalan C K, Sharma J, et al. Effect of Fault Ride Through Capability on Electric Vehicle Charging Station Under Critical Voltage Conditions. *IEEE transactions on transportation electrification*, vol. 2022, no. 2, pp. 8, 2022.
- [10] Nishanth J, Raja S C, Praveen T, et al. Techno-economic analysis of a hybrid solar wind electric vehicle charging station in highway roads. *International journal of energy research*, vol. 2022, no. 6, pp. 46, 2022.
- [11] Ye Z, Gao Y, Yu N. Learning to Operate an Electric Vehicle Charging Station Considering Vehicle-Grid Integration. *IEEE transactions on smart grid*, vol. 2022, no. 4, pp. 13, 2022.
- [12] Guillet M, Hiermann G, Krller A, et al. Electric Vehicle Charging Station Search in Stochastic Environments. *Transportation science*, vol. 2022, no. 2, pp. 56, 2022.
- [13] Zhu J, Li Y, Yang J, et al. Planning of electric vehicle charging station based on queuing theory. *The Journal of Engineering*, vol. 2017, no. 13, pp. 1867-1871, 2017.